Reinforcement Learning for Hydrothermal Scheduling

Comparison with SDDP

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Course Project: Stochastic & Dynamic Programming

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Motivation: SDDP vs. Reinforcement Learning

Stochastic Dual Dynamic Programming (SDDP)

- **Pros:** Mathematically exact, strict adherence to physical constraints, provides optimality bounds.
- **Cons:** Curse of dimensionality; increasing state-space drastically slows down convergence.

Reinforcement Learning (A2C)

- **Pros:** Highly scalable to large continuous state spaces.
- **Cons:** Struggles to strictly satisfy complex physical constraints (e.g., load balance), sample insufficient.

The Model: Hydro-Thermal Scheduling

- **System:** 4 Interconnected Subsystems (SE, S, NE, N).
- **Objective:** Minimize total expected operation cost over a 24-month horizon.
- **Cost Components:**
  - Thermal Generation Costs (95 dispatchable units).
  - Deficit Penalties (to penalize load shedding).
  - Exchange Costs (Transmission routing between regions).
  - Spill Costs (Small penalties to avoid numerical degeneracy).
- **Hydrology:** Log-normal inflows modeled as a Periodic Autoregressive PAR(1) process.

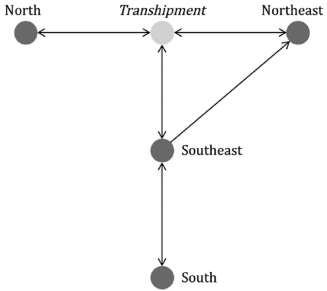


Figure: Interconnected power system

The Model: Hydro-Thermal Scheduling (Formulation)

Objective: Minimize expected operation cost over horizon T:

$$\min \mathbb{E}\left[\sum_{t=1}^{T}\left(\sum_{r,j} C_{r,j}^{th} g_{r,j,t} + \sum_{r,d} C_d^{def} z_{r,d,t} + \sum_{i,j} C_{i,j}^{exc} e_{i,j,t} + \epsilon \sum_{r} s_{r,t}\right)\right]$$

Subject to (per stage t):

- **Load Balance** (Physical Regions $r \in \{1, \ldots, 4\}$):

$$h_{r,t} + \sum_{j} g_{r,j,t} + \sum_{d} z_{r,d,t} + \sum_{i=1}^{5} e_{i,r,t} - \sum_{j=1}^{5} e_{r,j,t} = D_{r,t}$$

- **Exchange Balance** (Interconnection Node $k = 5$):

$$\sum_{i=1}^{4} e_{i,5,t} - \sum_{j=1}^{4} e_{5,j,t} = 0$$

- **Water Balance** (Regions $r \in \{1, \ldots, 4\}$):

$$v_{r,t} - v_{r,t-1} + h_{r,t} + s_{r,t} = I_{r,t}$$

- **Operational Bounds:**

$$0 \le v_{r,t} \le \overline{V}_r, \quad 0 \le h_{r,t} \le \overline{H}_r, \quad 0 \le g_{r,j,t} \le \overline{G}_{r,j}, \quad 0 \le e_{i,r,t} \le \overline{E}_{i,r,t}, \quad 0 \le z_{r,d,t} \le \overline{Z}_{r,d,t}$$

Variables: h (hydro), g (thermal), z (deficit), e (exchange), v (storage), s (spill), I (inflow state), D (demand).

Implementation 1: Time-Series SDDP (TS-SDDP)

Following the framework by *Löhndorf & Shapiro (2019)*:

- **State Space:** 8-dimensional continuous state space.
  - 4 Storage States (v_t)
  - 4 Inflow States (I_t) to restore stagewise independence.
- **Affine Transformation:** PAR(1) multiplicative noise mapped to affine constraints to preserve convexity.

$$I_t = \text{noise} \cdot (\eta_t + \Phi_t I_{t-1})$$

Implementation 2: Advantage Actor-Critic (A2C)

Architecture: Two neural networks (Actor θ, Critic ϕ), each with two hidden layers of 256 units.

State Space $s_t \in \mathbb{R}^{13}$ (normalized):

Component	Dimension
Month	1
Storage $v_{r,t}$	4
Inflow $I_{r,t}$	4
Demand $D_{r,t}$	4

Action Space $a_t \in \mathbb{R}^{144}$ — mirrors SDDP decision variables exactly:

Variable	Description	Dim
$h_{r,t}$	Hydro generation	4
$s_{r,t}$	Spill	4
$g_{r,j,t}$	Thermal generation (per unit)	95
$z_{r,d,t}$	Deficit (4 tiers $\times$ 4 regions)	16
$e_{i,j,t}$	Exchange (5 $\times$ 5 matrix)	25
	Total	**144**

Integrating SDDP and A2C

- **The Challenge:** A2C requires massive exploration to understand the physics.
- **The Solution: Imitation Learning**
  1. Extract the 500 training trajectories generated by SDDP (Expert Dataset).
  2. Supervised pre-training of the **Actor network** to map states to SDDP optimal actions (MSE loss).
  3. Supervised pre-training of the **Critic network** to map states to SDDP future Bellman costs.
- **Gradual transition:** mixed loss $(1-\lambda)\cdot\mathcal{L}_{\text{A2C}}+\lambda\cdot\mathcal{L}_{\text{imitation}}$ with λ annealing from $0.5\rightarrow 0$, so the agent continuously balances exploiting SDDP knowledge and exploring new policies ($\mathcal{L}$ is a loss function).

Implementation 2: A2C Dispatch Logic

Projection Pipeline — network outputs $\pi_\theta(s_t)$, projected to feasibility:

1. **Sigmoid scaling:** $a = \sigma(\pi_\theta(s_t)) \in (0,1)^{144}$, scaled to variable bounds
2. **Water balance (exact):** $s_{r,t} = \max(v_{r,t} + I_{r,t} - h_{r,t} - \overline{V}_r,\ 0) + s^{\text{opt}}_{r,t}$, $v_{r,t+1} = v_{r,t} + I_{r,t} - h_{r,t} - s_{r,t}$
3. **Load balance (approximate):** proportional scaling of thermal and deficit to close demand gap

Reward signal: $r_t = -\text{cost}_t$, where

$$\text{cost}_t = \sum_{r,j} C^{th}_{r,j} g_{r,j,t} + \sum_{r,d} C^{def}_d z_{r,d,t} + \sum_{i,j} C^{exc}_{ij} e_{i,j,t} + \epsilon \sum_r s_{r,t}$$

Computational Results: SDDP vs. A2C (with Imitation Learning)

TS-SDDP

Metric	Value
Iterations	1,000
State dimensions	8
Time	$\approx$ 25.2 mins (1512 s)
Best lower bound	54.80 M R$

A2C + Imitation Learning

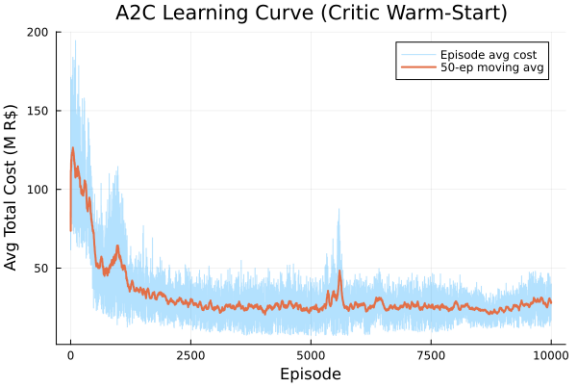
Metric	Value
Training episodes	10,000
State dimensions	13
BC warm-start	500 SDDP sims
Time	4 mins

Out-of-sample evaluation (5 random seeds, 100 scenarios each, mean cost and std):

	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
SDDP (M R$)	73.3 (69)	73.9 (70)	76.4 (85)	67.9 (53)	65.0 (50)
A2C (M R$)	33.7 (13)	34.2 (14)	36.9 (14)	36.5 (12)	34.7 (11)

Note: Water balance is exactly satisfied. Load balance is approximately satisfied via proportional projection; exact enforcement requires an embedded LP solver.

A2C Learning Curve



The descending learning curve demonstrates that imitation learning from 500 SDDP trajectories successfully warm-started the A2C policy. It eliminates the costly random exploration phase of RL from scratch.

Conclusion

Key Findings:

- **TS-SDDP** achieved a simulated expected cost of **73–76 M R$**.
- **A2C with imitation learning** achieved **34–37 M R$**,lower reported cost than SDDP, but not directly comparable due to approximate load balance enforcement.
- **A2C shows significantly lower variance** across seeds (std $\approx$ 12–14 M R$) compared to SDDP (std $\approx$ 69–85 M R$), suggesting the imitation-learned policy is more robust to different inflow scenarios.
- **Imitation learning** from 500 SDDP trajectories stabilized A2C training immediately, with the learning curve remaining flat around 25–35 M R$ throughout 10,000 episodes.
- **Water balance** is satisfied *exactly* by construction. **Load balance** is satisfied only *approximately* via proportional projection: the key limitation of A2C.